Practice workbook · The TCP/IP model: four layers and what each does

Naming the four layers (application, transport, network, data link) and stating what each does to packets on sending and receiving.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T56-04

1. Question 3 · part b · 4 marks

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Worked-example retry · Attempt part b only.

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3 Carlo uses his smartphone to make hands-free calls from the car.

(a) The smartphone connects to the car wirelessly forming a network.

Complete the table by adding a tick (✓) to indicate the network type and a tick (✓) to indicate the usage model the network uses.

(2)

Network type	Tick (✓)	Usage model	Tick (✓)
Local Area Network (LAN)		Client-Server	
Wide Area Network (WAN)		Peer-to-peer	
Personal Area Network (PAN)			

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(b) Data on a network is transmitted between devices using packets and the TCP/IP layered protocol stack.

Complete the table by naming **two** layers of the stack and giving **one** function for each of the layers that you have named.

(4)

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	Layer name	Function
1		
2		

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1. Question 3 · part b · 4 marks

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Worked-example retry · Attempt part b only.

(c) The smartphone can communicate using Wi-Fi.

Identify which **one** of these is a radio frequency used by smartphones to connect to Wi-Fi.

(1)

- ☒ **A** 2.4 GHz
- ☒ **B** 3 KHz
- ☒ **C** 4.1 GHz
- ☒ **D** 5 KHz

(d) Carlo's smartphone can use 3G, 4G and 5G mobile communication standards.

(i) Give **one** advantage of using a higher frequency band for mobile communications.

(1)

(ii) Explain **one** benefit to Carlo of using 5G rather than 3G.

(2)

(e) Carlo wants to reduce the environmental impact of his smartphone.

(i) Give **one** way Carlo could reduce energy consumption.

(1)

(ii) Explain **one** way, other than reducing energy consumption, that Carlo could reduce the environmental impact of his smartphone.

(2)

(Total for Question 3 = 13 marks)

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1. Question 3 · part b · 4 marks · Working and answer

Working and answer

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2. Question 1 · part f.ii · 2 marks

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Further practice · Attempt part f.ii only.

1 Upeka manages the network in her office building.

(a) Upeka uses the network to install software updates on all the office computers.

Give **two other** reasons why computers are connected in a network.

(2)

1 .....

.....

2 .....

.....

(b) The network uses wired and wireless connectivity.

(i) State **two** advantages of using wired rather than wireless connectivity.

(2)

1 .....

.....

2 .....

.....

(ii) State **two** disadvantages of using wired rather than wireless connectivity.

(2)

1 .....

.....

2 .....

.....

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2

P 6 1 8 8 4 R R A 0 2 2 0

2. Question 1 · part f.ii · 2 marks

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Further practice · Attempt part f.ii only.

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(c) Some people confuse the terms 'internet' and 'world wide web'.

Describe the difference between the internet and the world wide web.

(2)

(d) Networks use both IPv4 and IPv6 addresses to identify connected devices.

Explain why IPv6 addressing was introduced.

(2)

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2. Question 1 · part f.ii · 2 marks

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Further practice · Attempt part f.ii only.

(e) Network layouts can be based upon a number of different network topologies.

Figure 1 shows a network topology.

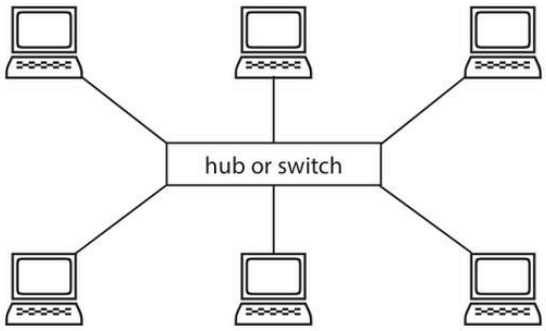


Figure 1

(i) State the name of this network topology.

(1)

(ii) State **one** benefit of using this network topology.

(1)

(iii) State **one** drawback of using this network topology.

(1)

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2. Question 1 · part f.ii · 2 marks

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Further practice · Attempt part f.ii only.

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(f) The network uses TCP/IP.

Figure 2 shows the TCP/IP protocol stack.

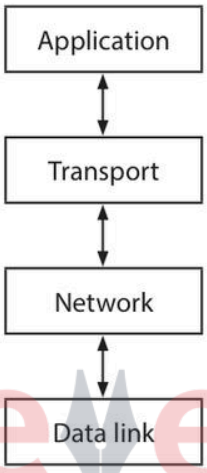


Figure 2

(i) One use of the application layer is to send and receive emails.

State **two** email protocols.

(2)

1 .....

2 .....

(ii) Upeka sends an email to Peter. The transport layer in Upeka's computer splits the email into data packets.

State **two** tasks performed by the transport layer in Peter's computer when the packets arrive from the network layer.

(2)

1 .....

.....

2 .....

.....

(Total for Question 1 = 17 marks)

2. Question 1 · part f.ii · 2 marks · Working and answer

Working and answer

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3. Question 2 · part c.i · 1 mark

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Further practice · Attempt part c.i only.

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2 Many organisations use networks.

(a) One reason for networking computers is to share access to the internet.

Give **one other** reason for connecting computers in a local area network.

(1)

.....

.....

(b) State the purpose of network protocols.

(1)

.....

.....

(c) Data is transmitted in packets.

(i) Identify the layer of the TCP/IP model that would add the File Transfer Protocol (FTP).

(1)

- ☒ **A** Application
- ☒ **B** Data
- ☒ **C** Network
- ☒ **D** Transport

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(ii) Give **one** task carried out by the network layer.

(1)

.....

.....

(d) Complete the table by adding a tick (✓) to match each item to its name.

(3)

Item	Domain Name	IP Address	URL
https://www.pearson.com			
216.33.251.195			
pearson.com			

3. Question 2 · part c.i · 1 mark · Working and answer

Working and answer

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4. Question 2 · part c.ii · 1 mark

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Further practice · Attempt part c.ii only.

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(1)

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https://www.pearson.com			
216.33.251.195			
pearson.com			

4. Question 2 · part c.ii · 1 mark · Working and answer

Working and answer

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5. Question 1 · part d · 1 mark

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Further practice · Attempt part d only.

1 There are different ways to connect computers in a network.

(a) Give **two** differences between a WAN and a LAN.

(2)

1 .....

.....

2 .....

.....

(b) Give **one** security risk of connecting computers in a network.

(1)

.....

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5. Question 1 · part d · 1 mark

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Further practice · Attempt part d only.

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(c) **Figure 1** shows a network topology.

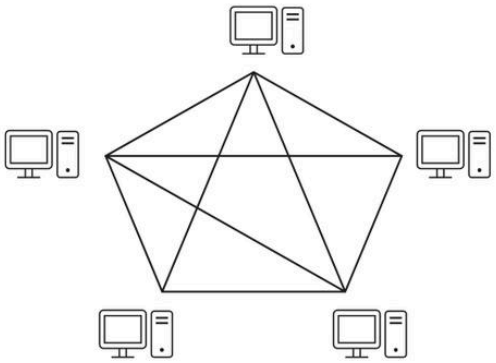


Figure 1

(i) State the name of this network topology.

(1)

(ii) State **one** drawback of using this network topology.

(1)

(iii) A fault tolerant network delivers a reliable service.

Explain **one** reason why this topology is fault tolerant.

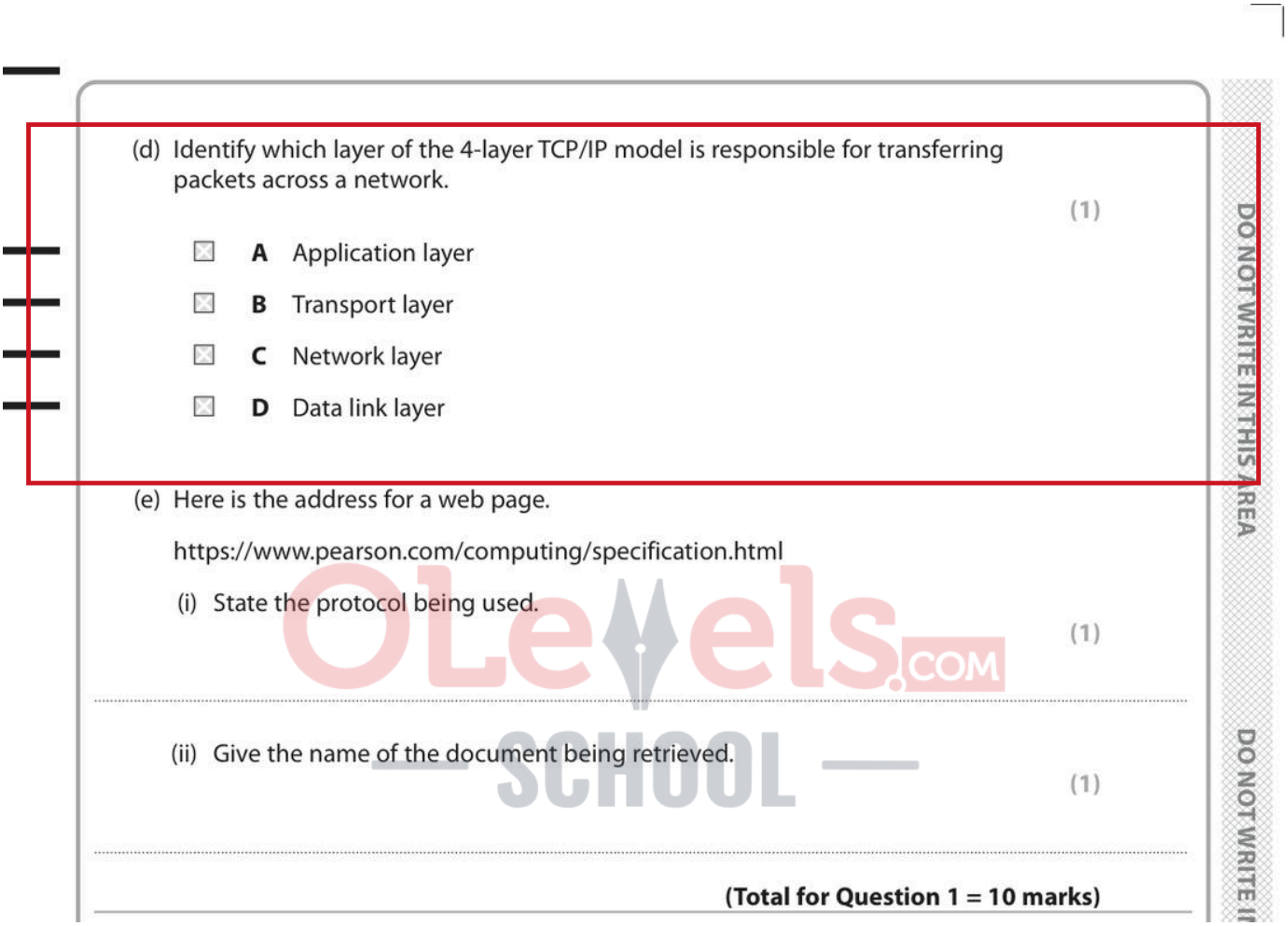
(2)

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5. Question 1 · part d · 1 mark

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Further practice · Attempt part d only.



(d) Identify which layer of the 4-layer TCP/IP model is responsible for transferring packets across a network.

(1)

- ☒ **A** Application layer
- ☒ **B** Transport layer
- ☒ **C** Network layer
- ☒ **D** Data link layer

(e) Here is the address for a web page.

https://www.pearson.com/computing/specification.html

(i) State the protocol being used.

(1)

(ii) Give the name of the document being retrieved.

(1)

(Total for Question 1 = 10 marks)

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Working and answer

6. Question 1 · part b · 2 marks

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Further practice · Attempt part b only.

1 Computers can be networked together.

(a) Complete the table by adding a cross ☒ to match **each** statement to **one** networking term.

(4)

Statement	DNS	PAN	WAN	WWW
A network of networks **not** limited to a single location	☒	☒	☒	☒
A network that connects devices within the range of a single user	☒	☒	☒	☒
A set of hypertext documents accessed through the HTTP protocol	☒	☒	☒	☒
A system that maps domain names to IP addresses	☒	☒	☒	☒

(b) Networks use the TCP/IP protocol model.

The application layer and the network layer are two layers in the TCP/IP model.

Give the names of the **other two** layers.

(2)

1 .....

.....

2 .....

.....

(c) Emails can be sent between computers on a network.

Identify the protocol required to **send** an email.

(1)

☒ **A** HTTP

☒ **B** IMAP

☒ **C** POP3

☒ **D** SMTP

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P 7 8 6 3 7 A 0 2 2 0

6. Question 1 · part b · 2 marks · Working and answer

Working and answer

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